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Experiments with Pintle Injector Design and Development

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This paper describes the results of an experimental investigation in designing, building and testing pintle injectors that are relevant to liquid rocket engine application. A team from UFMG was assigned the challenge to investigate the influence of some constraints on defining a pintle injector architecture that would be relevant in order to meet certain specifications, with focus on controlling the propellant mass flow. On the first part of this paper an overview of the pintle injector main characteristics is introduced, while on the second part, actual results obtained from the prototypes built are presented.

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Nomenclature

A	=	area
m	=	mass flow
C_d	=	discharge coefficient
ρ	=	density
P	=	pressure
Dc	=	chamber diameter
Dp	=	pintle diameter
Ls	=	skip distance

I. Introduction

Whenever a pintle injector related discussion takes place, TRW, now part of Northrop Grumman, is one of the first names mentioned. They deserve this recognition not only for the pioneer work started in the 60's, that eventually led to a patent, but also because they have employed pintle injectors on hundreds of engines[5], ranging from a 10-Newton thruster up to the experimental 3-mega-Newton Low Cost Pintle Engine (LCPE)[8]. Drawings, extracted from the patent filled by G.W. Elverum and TRW[7], are shown on Fig. 1. The original document describes the drawing on the left as "a perspective view, partially in cross section, of a coaxial injector" while the drawing on the right is described as "a view in cross section of an injector".

By the middle of the 60's, independently from TRW work, but in a much more modest scale, the Brazilian Space Research Institute-INPE developed several versions of a 200 N hypergolic rocket engine[6], which incorporated what can now be identified as a coaxial pintle injector.

The latest news about pintle injector application comes from Space-X, a company that is changing the space business by making commercial launch vehicles a reality. According to Ref. 11, Space-X Falcon rocket was designed by a former TRW engineer who may have his professional experience incorporated into the design of the Merlin and Kestrel engines, including the application of a central pintle injector.

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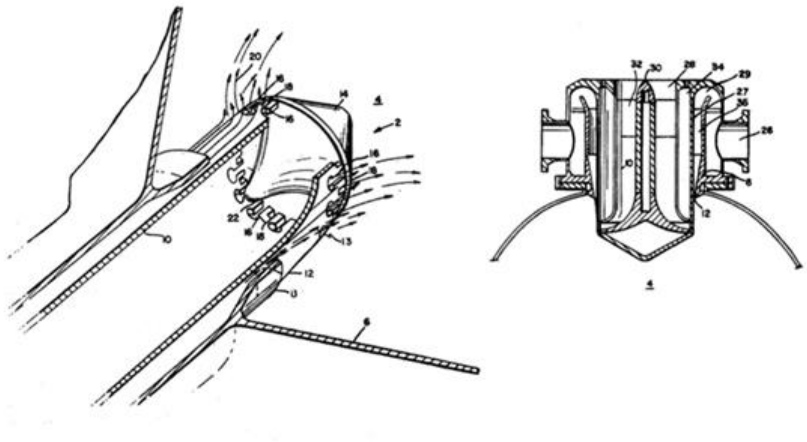

Figure 1 – Co-axial pintle injector as per Patent 4,206,594[7].

When compared with the swirl injector, there is not much information on pintle injector characteristics available in the open literature, or at least in the data base our team had access to. The situation is aggravated by limited academic studies, design codes kept for the industry use only and the little attention given to the physics associated with the overall impingement characteristics[1]. It was under these circumstances that a team was put together, at UFMG- Minas Gerais Federal University, to investigate the constraints used on designing a pintle injector and to explore cost effective fabrication options that would be relevant to create a product that could meet certain specifications, more specifically to deliver the required amount of propellant. This paper will describe the challenges faced by the team, as well as provide an in depth analysis of the solutions found, including details on the injector internal arrangements. Figure 2 contains pictures of the three injectors mentioned on this work, starting with a proof of concept model designed for a hypothetical 500 N engine that provided some interesting information on key elements requiring extra attention. Then, making good use of the lessons learned it was possible to develop the second prototype, for a 1 kN engine. It was followed by the third one, our largest so far, the 5 kN capable injector, that incorporates some valuable skills acquired by the designing team in creating a modular and scalable product, adaptable to many different operational conditions with minor modifications.

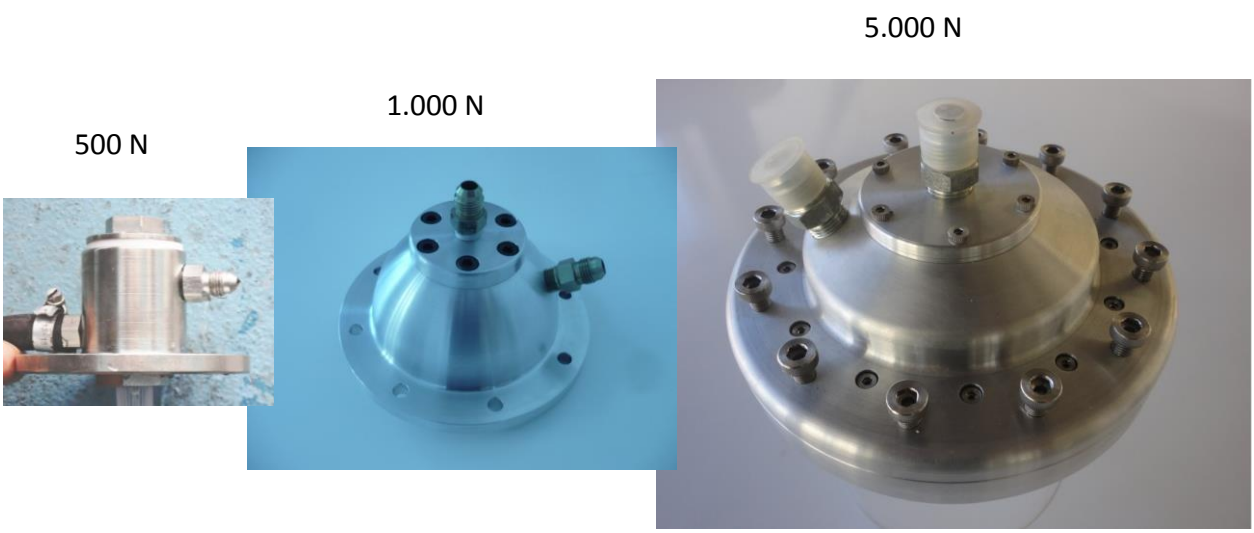

Figure 2 – Pintle injectors designed by the UFMG team.

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II. Injector Overview

The pintle injector has demonstrated its effectiveness in atomizing and mixing propellants in rocket engines, associated with some performance degradation in the range of 1% to 4%, when compared with the theoretical values[5]. Another interesting aspect is the pintle injector inherent protection against combustion instability, a conclusion driven by the observation that TRW has never experienced combustion instability in any of its pintle injector equipped engine[12]. On the manufacturing side, the main advantage of the pintle injector is its relative simplicity when compared with injectors of other configurations. It can be designed to incorporate the possibility for quickly changes on the critical injection elements, while keeping the main body unaffected. Parts counting can be reduced to a bare minimum, with the vast majority of them built as solid of revolution, which means they can be manufactured using ordinary industrial grade machining centers. However, special efforts are still needed in order to guarantee the metering orifices dimensions accuracy or the correct size of the annular gap controlling the axial flow.

On a typical pintle injector, as a version shown on Fig. 3, one of the propellants enters the device from the upper part, flows down inside the pintle and is ejected through a set of calibrated orifices located at the pintle tip, which delivers metered propellant mass flow into the combustion chamber, in the form of radial lines much like a bicycle wheel spoke. The other propellant enters the injector laterally and is directed into a toroidal chamber, pass through a multi role orifice plate responsible for distributing the propellant evenly across the lower chamber, before reaching the combustion chamber in the format of a cylindrical thin sheet attached to the surface of the pintle. Collisions between the radial and the axial flow stream result in the appropriated mixing and atomization. Usually, but not necessarily, fuel is injected in the inner part (radial) and oxidizer on the external part (axial).

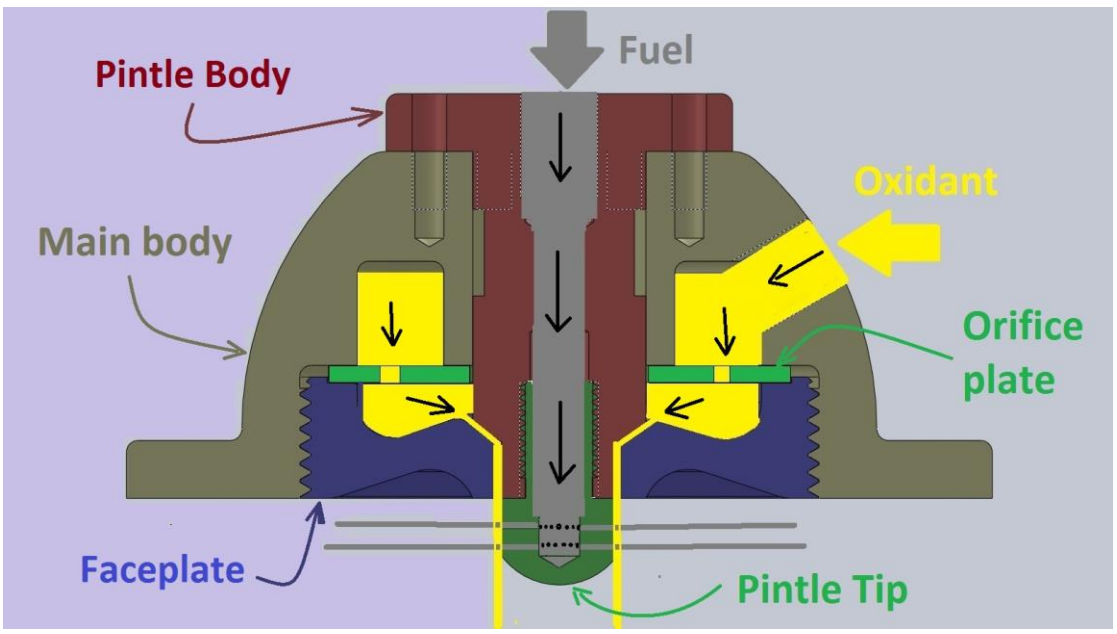

Figure 3 - Elements of a typical pintle injector.

During the initial experimental phase, several candidate configurations were tried, some with more success than others, until a decision was made to concentrate efforts into investigating the potential of two options for the axial flow, two options for the radial flow and the possible combination among them.

Control of the axial flow was achieved with a continuous gap ring that produces a hollow cylinder as per Fig. 4a, and a toothed gear like ring that creates a hollow cylinder defined by several lines of propellant, shown on Fig. 4b.

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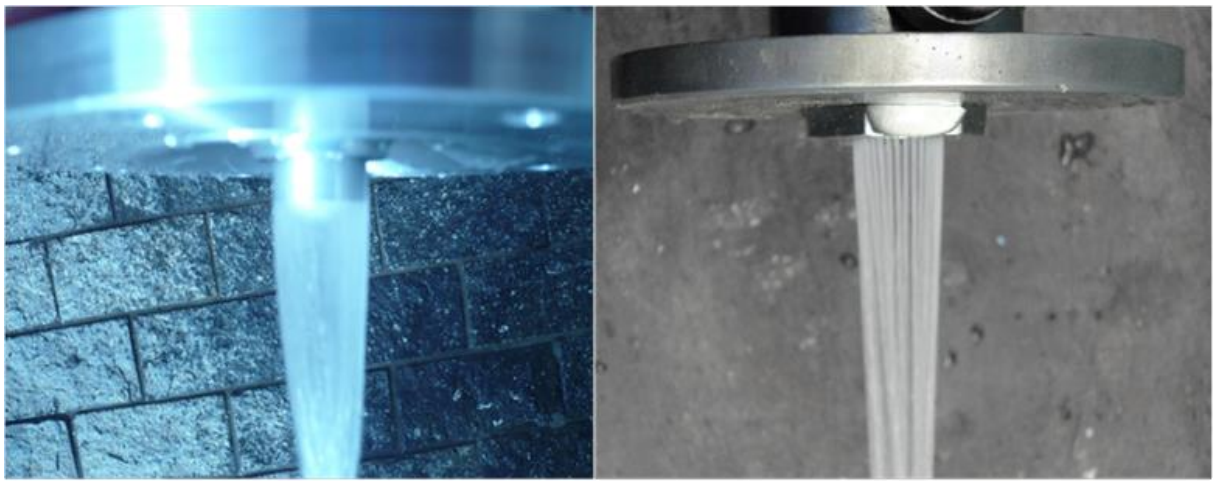
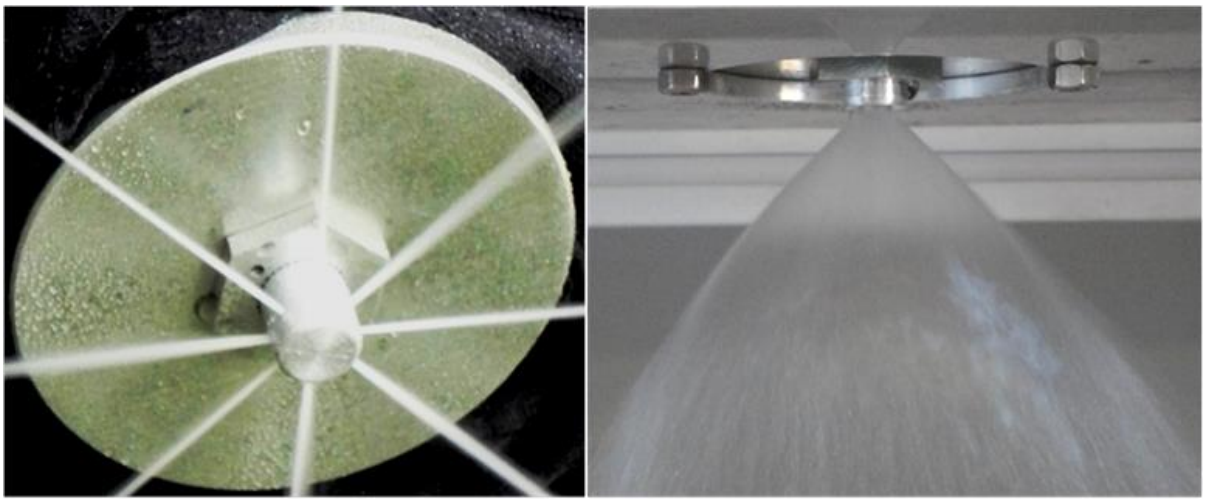


a) Continuous Axial Flow b) Discrete Axial Flow

Figure 4– Continuous and Discrete Axial flow

To control the radial flow, two pintle tip configurations were considered: one with multiple radial jets as the picture on Fig. 5a, and another incorporating a swirl injector, as presented on Fig. 5b.



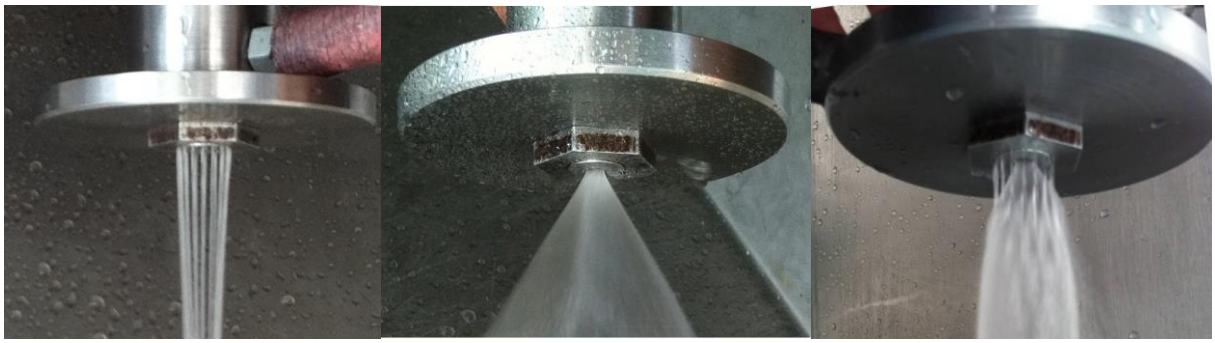
a) Multiple Radial JetsFlow b) Swirl Radial Flow

Figure 5 - Multiple Jets and Swirl Radial flow

Selection of a particular combination of radial and axial injection among the four suggested patterns is driven primarily by the mass flow required for proper operation, including flow required for chamber wall film cooling, associated with the manufacturing process available to achieve the specifications. As a general rule, it can be said that the continuous axial flow from Fig. 4a, combined with the multiple jet radial flow on Fig. 5a, is the preferred pair for most application, particularly with large engines. For smaller engines, something under 1 kN, when the orifice sizes are small enough to create difficulties with precision drilling, our option of choice goes to swirl radial flow, as per Fig. 5b, combined with either continuous or discrete axial flow, as per Fig. 4. A no go option for any engine size would be the combination of discrete axial and jet radial flow, because there is always the possibility that the jets don't impinge on each other.

An example of the mixing and atomization resulting from the collision of the axial jets with the thin, rotating liquid sheet of the swirl radial injector is shown on Fig.6. From the sequence of photographs taken during the water flow tests of the 500 N injector, Fig. 6a shows the water jets flowing axially through the toothed gear gap, while on

Fig. 6b is shown the radial swirl flow through the inner part of the pintle. Then, the picture on Fig. 6c shows the combination of the two flows and the resulting fan spray.



a) Axial flow only b) Radial flow only c) Combination

Figure 6 - Multiple Jets Radial and Swirl Axial flow combination

III. Design Practices

The pintle injector here presented can be described as a hydro-mechanical propellant control unit that delivers a fixed amount of propellant to the engine combustion chamber, by maintaining a constant pressure drop across an array of fixed cross section area metering tubes. Some degree of variation on the propellant mass flow is possible through adjustments on the inlet pressure, but no attempt was made to develop a throttling device. The main metering function was split between the pintle tip and the annular gap existing between the pintle body and the faceplate, which have an area dependent on a number of factors required to guarantee the desired propellant mass flow. One version of a pintle injector is shown on Fig. 7, with a section view and its respective exploded view. This configuration is flexible enough to accept adjustments on propellant mass flow after minor modifications on hardware, either by replacing the pintle tip or adjusting the annular gap, with no need to change the injector body

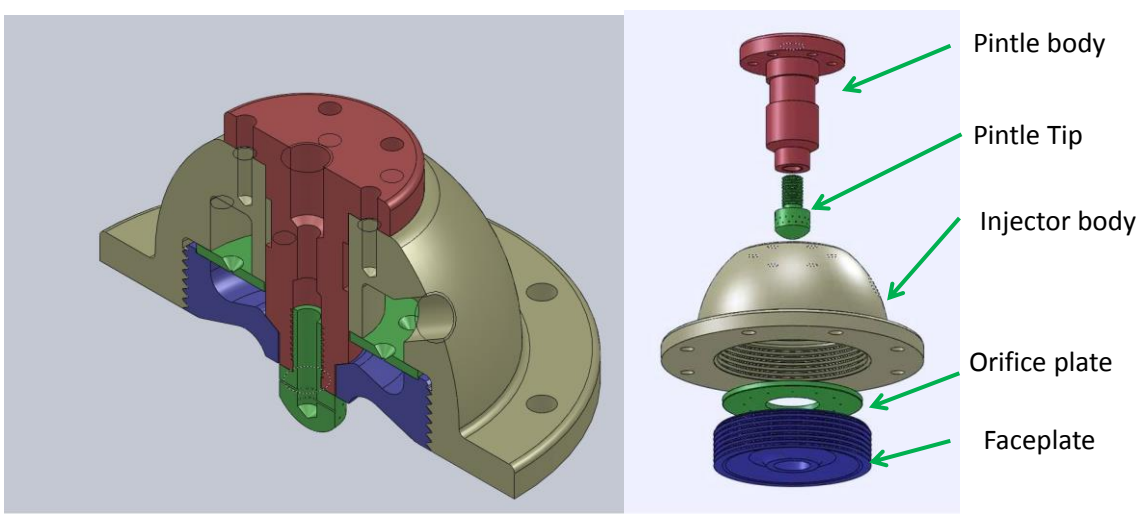


Figure 7 – Pintle injector – Section and exploded view.

Definition of the pintle diameter and how deep it is positioned into the combustion chamber is based on the required length to provide reaction time and stability, the resultant propellant film thickness and the pintle circumference necessary to locate the injection orifices[4]. The pintle can be related to the combustion chamber diameter on the proportion of 1/3 to 1/5 of the chamber diameter, with the resulting circumference defined from the perspective of the length needed for locating the metering holes.

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The distance between the injector face and the first line of holes, called skip distance, can be estimated as equal to the diameter of the pintle[1]. Figure 8 shows a diagram of some of the key elements required for definition of the pintle dimensions and location.

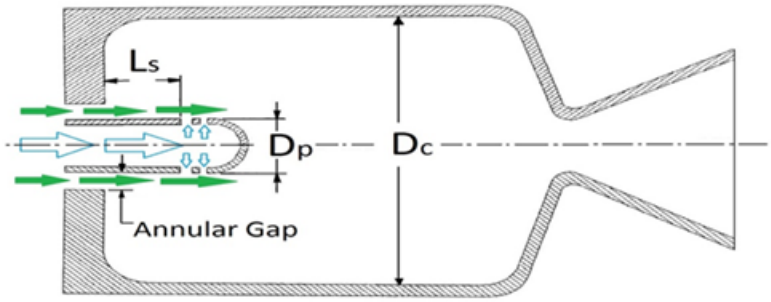

Figure 8 – Pintle elements[1]

Besides defining the internal architecture of the pintle, another important objective of this work was the understanding of how the metering tubes cross section affects the delivery of propellants.

Consider that the hydraulic flow of an incompressible fluid through an orifice is defined by Eq. (1).

$$A = \frac{\dot{m}}{C_d \cdot \sqrt{2 \cdot \rho \cdot \Delta P}} \qquad (1)$$

Where, A (m^2) is the total area of the orifices used to meter propellants flow, $\dot{m}$ (kg/s) is the propellant mass flow, C_d (dimensionless) is a discharge coefficient, ρ (Kg/m^3) is the fluid density and ΔP (Pa) is the pressure differential between the injector inlet pressure and the combustion chamber pressure. In order to proceed with the calculations, that eventually will get the orifice area, a set of information has to be provided by the customer (basically, m, ρ, and ΔP), leaving the discharge coefficient (C_d) to be defined by the designing team.

On small rocket engines, when the injector requires small flow area passage, one option to dodge the difficulties of dealing with small diameters holes drilling accuracy is using a radial swirl injector central pintle as the one shown on Fig. 5b. In general, for the same mass flow swirl injectors have lower discharge coefficient than jet injectors do, which translates in larger flow passage area. On top of that we have fewer, relatively larger holes, instead of several small diameter holes as on jet injectors. Defining the discharge coefficient for the swirl injector requires matching the dimensions of the inlet tangential orifices with the dimensions of the outlet orifice and the spray angle, creating a rather complex environment. One solution for this situation is presented on Ref. (9), showing that for a given spray angle, the exit and the inlet discharge coefficient although not equal, are constant across a wide range. For example, the 90 degree spray cone angle injector, operating with a pressure differential of 3.0 Bar, has an exit discharge coefficient of 0.260 and the inlet discharge coefficient is 0.319.

On the other hand, defining the discharge coefficient for jet injectors, with configuration similar to those shown on Fig. 5a, is a relatively straight forward procedure, considering a few conditions are met. The first one is to characterize the metering hole as a short tube with sharp edge entrance[2]. This condition can be achieved when using a sharp angle at the inlet edge by making β = 0, as per the definition on the left side of Fig. 9, and providing a tube with a length of at least 2 times its diameter (Li = 2 x di), so allowing sufficient distance for the expanding jet to reach the injector wall, eliminating the contraction effect. Using the sketch on the right side of Fig. 9 it can be noticed that the hole's length is a good parameter to define the minimum thickness of the orifice plate or the pintle wall minimum thickness. Applying these concepts to the design procedures defined on Ref. (2), a preliminary evaluation places the discharge coefficient in 0.70 to 0.80 range. One important missing factor from this procedure is the influence of the manufacturing process, as wall surface roughness plays a crucial role in defining frictional losses and the ability to consistently drill accurate holes influences the flow rate.

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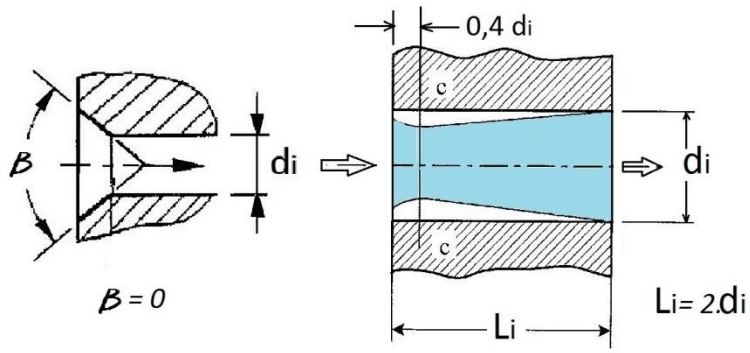

Figure 9 – Sketches for the jet injector[2].

To confirm the above mentioned assumptions and to incorporate the manufacturing process induced variables, a test specimen was designed to work with a hypothetical 500 N engine. It has a multiple jet radial flow pintle tip, similar to that shown on Fig. 5, with eight 1 mm diameter holes, giving a total discharge area of 6.28 mm^2. It was connected to a test bench where data related to inlet pressure and mass flow was collected. Actual data from the test bench is shown on Fig. 10, where the blue round dots represent the water mass flow while the continuous green curve represents the trend line. Next to it is an equation that relates the rate flow to the inlet pressure. Mass flow varies from 73 g/s at 1.0 Bar, to 125 g/s at 4.5 Bar, reaching 106 g/s at 3.0 Bar. The discharge coefficient is plotted as the red dotted line, presenting decreasing values from 0.83 to 0.66 as water mass flow and inlet pressure increase. Data derived from the test allows the conclusion that the speed of the water jet at the moment it leaves the metering tube increases with pressure, from 12 m/s to 20 m/s, reaching 17 m/s when the inlet pressure reaches 3 Bar.

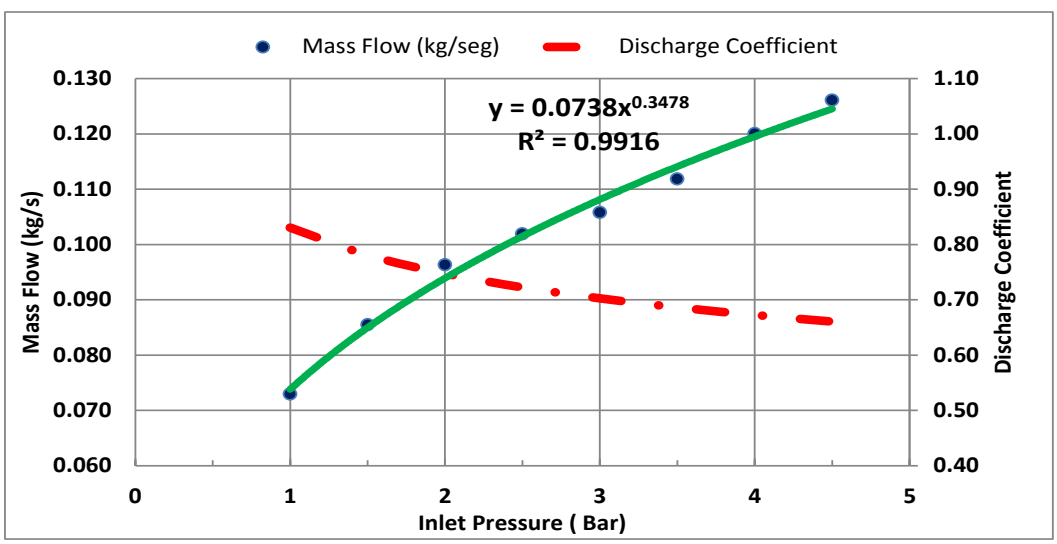

Figure 10 – Data from a test specimen

IV. Case Study – The 1 kN injector

Based on the experience accumulated with the first test specimen, a proof of concept, scaled up version of the coaxial flow injector was designed, built and cold tested. Figure 11 shows the details of the injector, designed to deliver the mass flow demanded by a 1 kN rocket engine, which is equivalent to 0.185 Kg/s of water through the metering plate (axial) and 0.125 kg/s through pintle tip (radial), under a constant inlet pressure of 3.0 Bar.

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Figure 11 – The 1 kN pintle injector parts breakdown

The first part of the test was conducted to determine flow rate, inlet pressure, mixture ratio and general impingement characteristics using water instead of the actual propellants. Figure 12 gives an overview of the test apparatus with the injector attached to hoses, digital manometers and flowmeters. Connection on the upper side represents the fuel line while the connection on the left represents the conduit for oxidizer. One interesting feature is the acrylic vessel simulating the combustion chamber that was installed with the objective of containing the water exiting the pintle tip within the limits of the acrylic camera, avoiding spreading water over the entire lab and allowing for a visual inspection of the water flowing out of the injector.

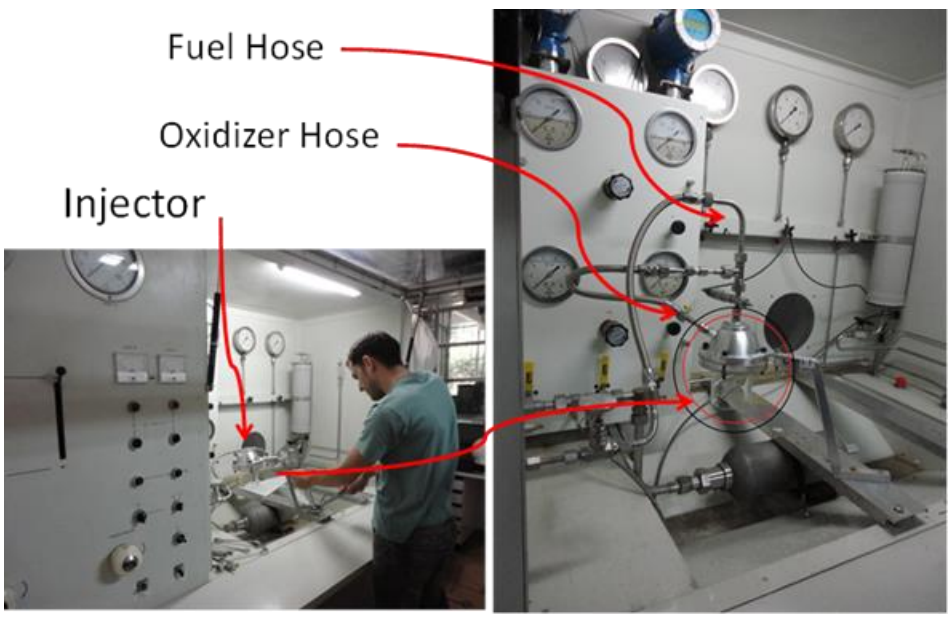
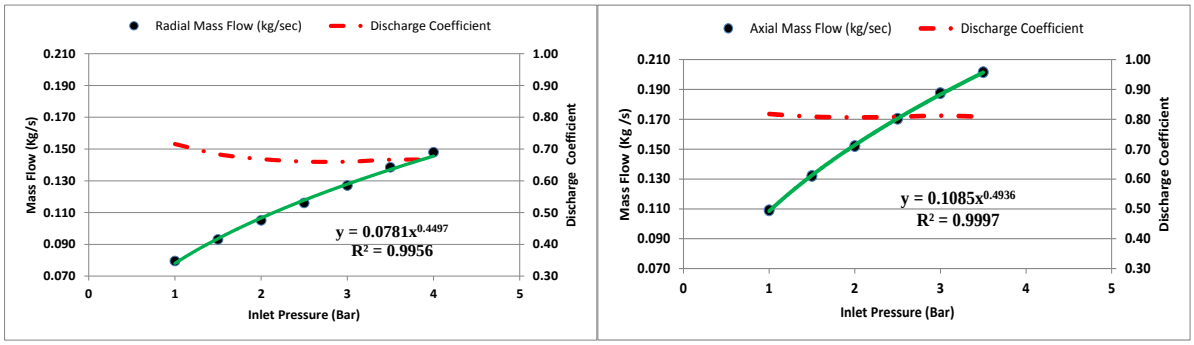


Figure 12 – At the test bench.

The mass flow rate for both, radial and axial flow, as a function of the inlet pressure was determined, with results showing on Fig. 13, including the respective injector equation. From this particular application we have that at 3.0 Bar the radial flow reached 128 g/sec, while the axial flow was 187 gr/sec, which is close enough to the target after the third iteration and within the error limits of the equipment available at the test bench. The discharge coefficients was 0.81 for axial flow and 0.67 for the radial flow, at 3.0 Bar. Options to avoid using a too expensive manufacturing process relies on the introducing minor adjustments on the mass flow by drilling a few extra

calibrated orifices or increasing the diameter of some holes as required to achieve the desired results, then, proceeding to the next round of test and calibration. On this case, it started with 32 holes on the pintle that was followed by the addition of four extra holes before satisfactory results were achieved.



a) Radial Flow chart b) Axial Flow chart

Figure 13 – Mass Flow as a function of the Inlet Pressure.

The overall impingement characteristics were visually verified, and recorded, with the aid of a digital camera, that filmed the event and took pictures like those shown on Fig. 14. Picture on the left shows the circumferential uniformity on the axial spray pattern flowing out of the injector in the format of a hollow cylinder, while picture on the right shows the radial flow hitting the acrylic cylinder and flowing down attached to its wall.

Figure 14 – Injector at work

V. Case Study – The 5 kN injector

With funding provided by a Brazilian government agency our team jumped into a new challenge, which required the development of a new injector for a 5 kN thrust engine that would have a consumption of 0.6 kg/s of ethanol and 1.0 kg/s of liquid oxygen under a pressure differential of 3.0 Bar. Another point of reference was that the injector assembly should match the already existing engine mounting plate. The injector architecture followed the multi layer configuration previously tried on the 1 kN injector, where components are stacked and bolted up together and have o ’rings seals in between layers. It also incorporates the flexibility to accommodate critical hardware modifications with minimum efforts, as well as, some mechanical solutions borrowed from Ref. 10.

The resulting product that emerged from the specifications, including the acrylic combustion chamber mock up, can be seen on Fig. 15.

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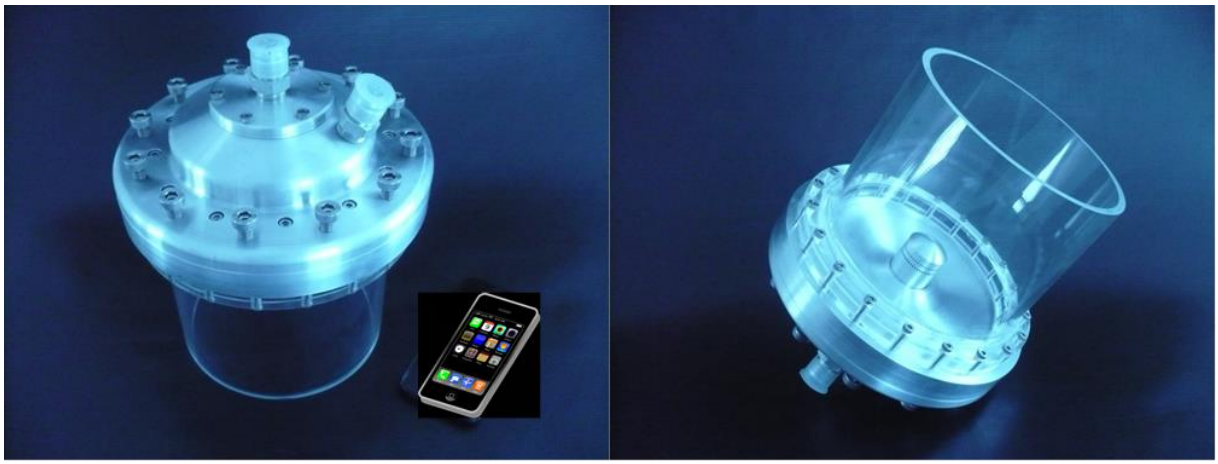

Figure 15 – The 5 kN pintle injector

The injector consists basically of three subassemblies, the injector body, the faceplate and the pintle. The pintle, that is responsible for fuel injection, was sub divided into three parts, the fuel inlet connector, the pintle body and the pintle tip, containing two equally spaced rows of 1 mm in diameter orifices. The primary row, located closer to injector face, is followed by a secondary row located next to the pintle tip. Replacing the threaded injector tip can modify the radial flow mass, velocity or geometry. For details refer to Fig. 16.

Oxidizer is fed into the injector by a lateral connection on the injector body , pass through a perforated plate that distribute the propellant into a second chamber before going through the annular gap between the pintle and the injector faceplate. Replacing shims at the top of the injector will change the annular passage dimensions, allowing adjustments to the oxidizer mass flow and velocity. All parts are kept together through a series of bolts. Materials used on this prototype were 304 stainless steel on the injector body and faceplate, and 6061 aluminum on the pintle, subject to modifications after the initial calibration test.

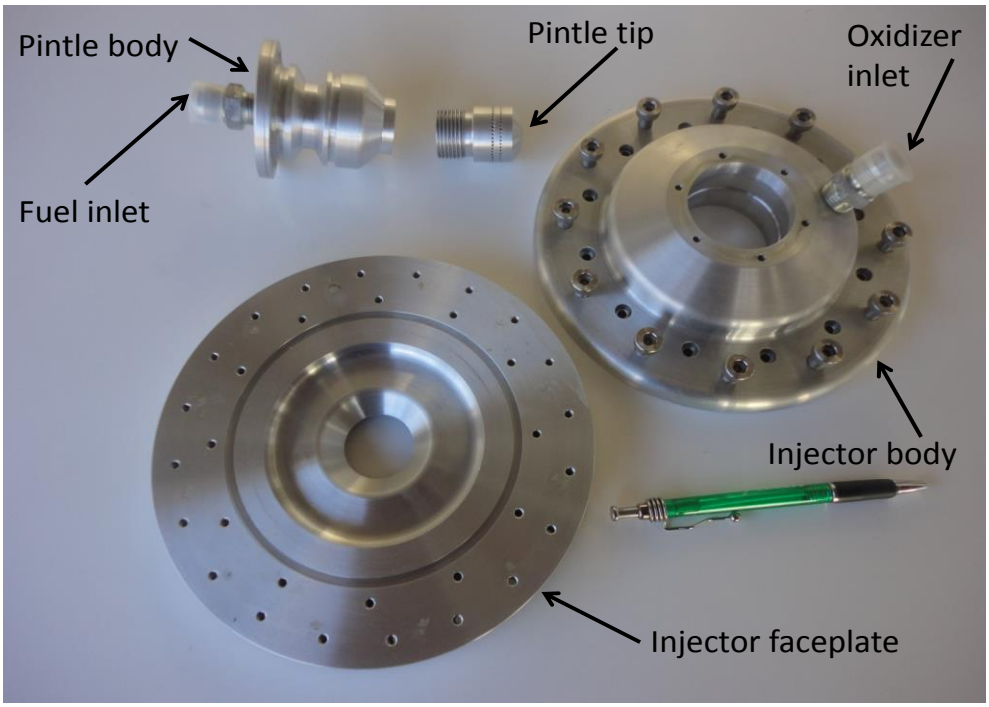

Figure 16 – The 5 kN pintle injector parts breakdown

VI. Conclusion

Simple architecture, relatively lower cost, inherent combustion stability and multiple opportunities for exploring fundamental problems associated with the pintle injector mixing and atomization make it an interesting option for academic studies. Under these assumptions, a team from UFMG developed a set of empirical procedures to design its version of the pintle injector, with focus on the procedures required to guarantee the delivery of the required mass flow. Test specimens have been fabricated and successfully tested in laboratory to verify the validity of these procedures. Actual and forecasted data were compared and found to be within reasonable limits, including the relationship between mass flow and pressure drop across the injector. In spite of that, it is clearly understood that there is still a long way to go before the studies are completed and an operational injector is available for hot test.

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